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Entity Relationship Diagram (ER Diagram) Overview

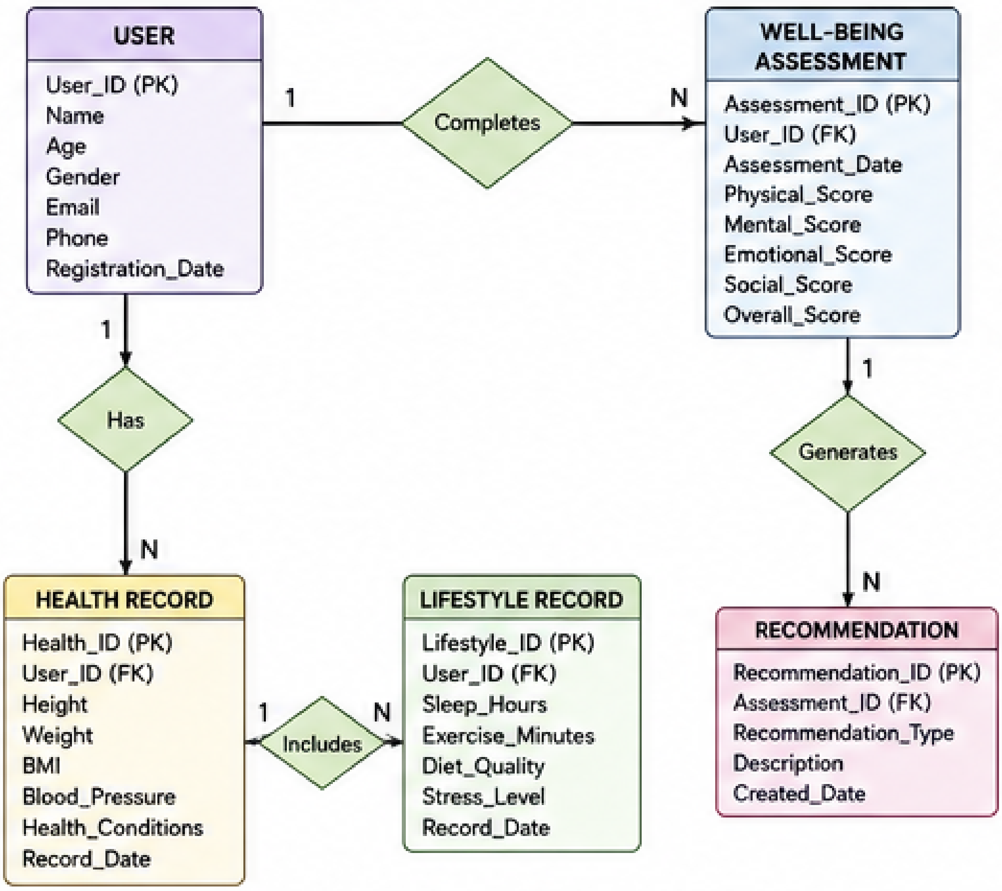
The Entity Relationship (ER) Diagram represents the database structure of the "A Comprehensive Measure of Well-Being" system. It illustrates the entities involved in the application, their attributes, and the relationships among them. The ER diagram helps developers understand how data is organized and connected.

The main entities in this project include User, Well-Being Assessment, Health Record, Lifestyle Record, and Recommendation. Each entity stores specific information related to the user's physical, mental, and emotional well-being.

Relationships between these entities ensure that user data is managed efficiently without redundancy. The ER diagram also supports database normalization and improves data integrity.

It serves as a blueprint for designing the database before implementation. Overall, the ER diagram provides a clear visualization of the system's data model.

Diagram 1: Overall ER Diagram



Entities and Their Relationships

The User entity stores personal information such as User ID, Name, Age, Gender, and Contact details.

Each user can complete multiple Well-Being Assessments, making it a one-to-many relationship.

Every assessment records scores related to physical health, mental health, emotional health, and social well-being.

The Health Record entity maintains medical information such as BMI, blood pressure, and health conditions.

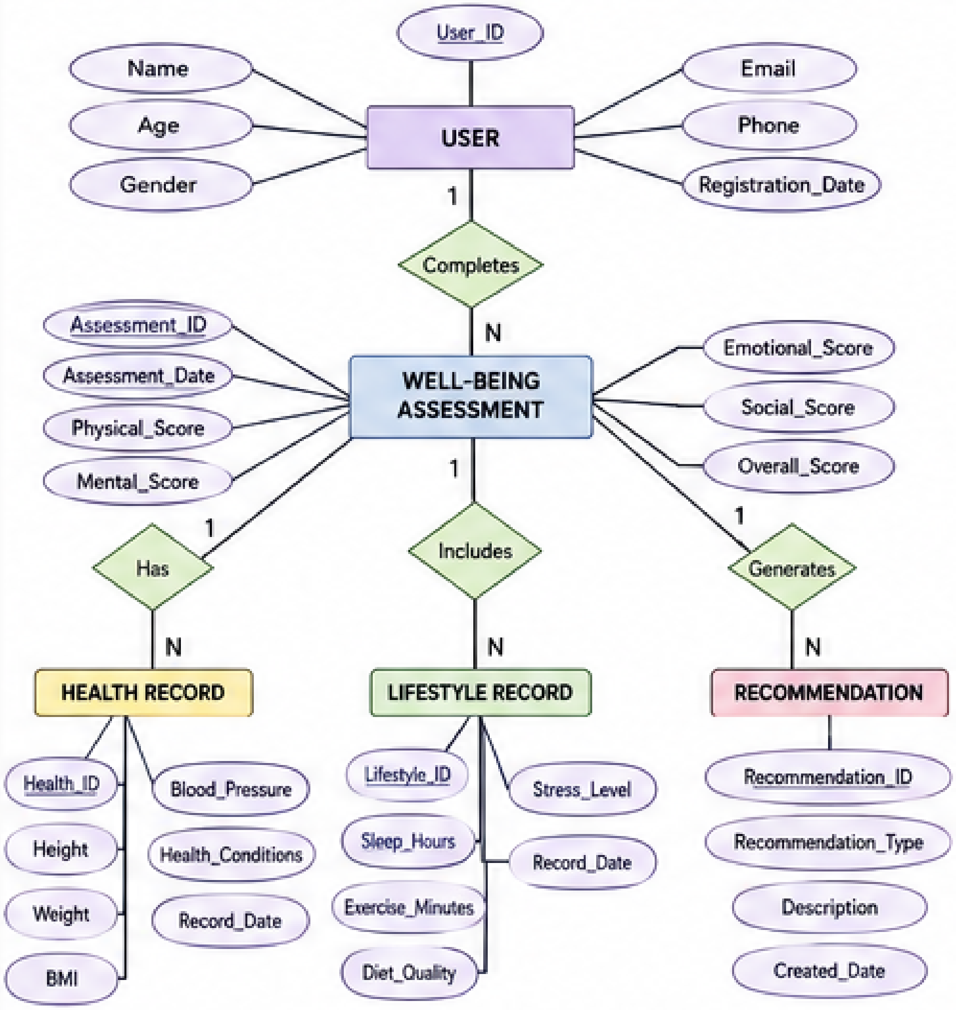
The Lifestyle Record entity stores daily habits including sleep duration, exercise, diet, and stress level.

Based on the collected data, the Recommendation entity provides personalized suggestions for improving overall well-being.

Each recommendation is linked to a specific assessment.

These relationships ensure efficient data retrieval and accurate analysis of user well-being.

Diagram 2: Detailed ER Diagram – Attributes View



Importance of the ER Diagram

The ER diagram simplifies database design by providing a visual representation of data relationships.

It helps identify the entities, attributes, and keys required for the system.

Proper relationships reduce data duplication and maintain consistency across the database.

The ER model improves communication between developers, database designers, and project stakeholders.

It also makes future modifications and maintenance easier.

During implementation, the ER diagram acts as a reference for creating database tables and defining primary and foreign keys.

A well-designed ER diagram enhances system performance and supports accurate data analysis.

In this project, it ensures reliable storage and management of well-being information, enabling effective monitoring and personalized recommendations for users.

Diagram 3: ER Diagram – Crow's Foot Notation